THE GROTHENDIECK SPECTRAL SEQUENCE

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Abstract. Given left exact functors F and G between abelian categories, the Grothendieck spectral sequence computes the right derived functors of the composite $G \circ F$ from those of F and G; it is the "chain rule" of homological algebra. We give a self-contained account of its construction. After reviewing derived functors and the spectral sequence of a filtered complex, we build the spectral sequence from a Cartan–Eilenberg resolution and extract the five-term exact sequence. We then specialize to the Leray, Lyndon–Hochschild–Serre, and change-of-rings spectral sequences, and use each to carry out an explicit computation.

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Date: October 14, 2026.

2020 *Mathematics Subject Classification.* Primary 18G40; Secondary 18G10, 55T05, 20J06, 55N30.

Key words and phrases. Spectral sequences, derived functors, Cartan–Eilenberg resolutions, Leray spectral sequence, group cohomology.

1. Introduction

Many functors in algebra and geometry are naturally composites. Global sections of a sheaf on X can be computed by first pushing the sheaf forward along a map $f\colon X \to Y$ and then taking global sections on Y; the invariants of a group Π can be computed by first taking invariants under a normal subgroup N and then under the quotient Π/N. In each case one would like to compute the derived functors of the composite from the derived functors of the pieces. In general there is no simple formula, but there is the next best thing: a spectral sequence

$$E_2^{p,q} = R^pG\big(R^qF(A)\big) \implies R^{p+q}(G\circ F)(A),$$

introduced by Grothendieck in his Tôhoku paper [Gro57]. It generalizes many spectral sequences that had previously been constructed by hand, including the Leray spectral sequence of a continuous map and the Hochschild–Serre spectral sequence in group cohomology, and it remains one of the basic tools in both subjects.

The aim of this note is to prove the existence of the Grothendieck spectral sequence carefully and to show it in action. Section 2 reviews abelian categories and right derived functors. Section 3 develops the spectral sequence of a filtered complex, with a complete proof, and specializes it to the two spectral sequences of a double complex. Section 4 introduces Cartan–Eilenberg resolutions and uses them to construct the Grothendieck spectral sequence. Section 5 derives three classical special cases, and Section 6 uses them to carry out explicit computations, including two in which the spectral sequence has nontrivial differentials.

Our treatment of filtered complexes follows [Lan02] and [Wei94], and the proof of the main theorem follows the outline in [Tor] and [Wei94, §5.8]. Throughout, "complex" means cochain complex: differentials raise degree by one.

2. Derived Functors

Derived functors measure the failure of a left exact functor to be exact. Familiar examples include Ext, group cohomology, and, most importantly for us, sheaf cohomology. We begin by recalling the setting in which they are defined.

2.1. Abelian categories.

Definition 2.1. A category $\mathcal{A}$ is *abelian* if

(1) each $\mathrm{Hom}(M,N)$ is an abelian group and composition is bilinear;
(2) $\mathcal{A}$ has a zero object and finite direct sums;
(3) every morphism has a kernel and a cokernel;
(4) every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel.

It follows from these axioms that every morphism factors as an epimorphism followed by a monomorphism, so images make sense, and hence so do exact sequences, complexes, and cohomology. Examples include the category **Ab** of abelian groups, the category Mod_R of modules over a ring R, the category $\mathrm{Sh}(X)$ of sheaves of abelian groups on a topological space X, and the category of sheaves of $\mathscr{O}_X$-modules on a ringed space $(X,\mathscr{O}_X)$.

For intuition one may think of an abelian category as a category of modules. This can be made precise: by the Freyd–Mitchell embedding theorem, every small abelian category admits a fully faithful exact functor into Mod_R for some ring R [Wei94, §1.6]. In practice this means that diagram chases with "elements" are legitimate in any abelian category.

Definition 2.2. A functor $F\colon \mathcal{A} \to \mathcal{B}$ between abelian categories is *additive* if each map $\mathrm{Hom}(A', A) \to \mathrm{Hom}(F(A'), F(A))$ is a homomorphism of abelian groups. An additive functor F is *left exact* if for every short exact sequence $0 \to M' \to M \to M'' \to 0$ the sequence

$$0 \to F(M') \to F(M) \to F(M'')$$

is exact; it is *right exact* if $F(M') \to F(M) \to F(M'') \to 0$ is exact; and it is *exact* if it is both.

A left exact functor generally does not preserve surjections. The right derived functors R^nF constructed below repair this by extending the truncated sequence above to a long exact sequence.

Additive functors automatically preserve finite direct sums, and hence preserve *split* short exact sequences. This simple observation is the engine of the proof of the main theorem.

2.2. Injective objects and derived functors.

Definition 2.3. An object I of $\mathcal{A}$ is *injective* if the contravariant functor $\mathrm{Hom}(-, I)\colon \mathcal{A} \to \mathbf{Ab}$ is exact. We say $\mathcal{A}$ *has enough injectives* if every object M admits a monomorphism $M \hookrightarrow I$ into an injective object.

The categories Mod_R, $\mathrm{Sh}(X)$, and $\mathrm{Mod}(\mathscr{O}_X)$ all have enough injectives [Wei94, §2.3], [Har77, III.2.2, III.2.3]. If $\mathcal{A}$ has enough injectives, every object M has an *injective resolution*: an exact sequence

$$0 \to M \to I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \to \cdots$$

with every I^n injective. We write $M \to I^\bullet$, where $I^\bullet$ denotes the complex $I^0 \to I^1 \to \cdots$ (with M removed).

From now on, $\mathcal{A}$ has enough injectives and $F\colon \mathcal{A} \to \mathcal{B}$ is additive and left exact.

Definition 2.4. Let $M \to I^\bullet$ be an injective resolution. The *n-th right derived functor* of F is

$$R^nF(M) = H^n\big(F(I^\bullet)\big),$$

the n-th cohomology of the complex $F(I^0) \xrightarrow{F(d^0)} F(I^1) \xrightarrow{F(d^1)} F(I^2) \to \cdots$.

Because F is left exact, applying F to $0 \to M \to I^0 \to I^1$ gives an exact sequence $0 \to F(M) \to F(I^0) \to F(I^1)$, so $R^0F(M) = \ker F(d^0) \cong F(M)$. The following theorem collects the basic properties; see [Har77, Theorem III.1.1A] or [Wei94, §2.4].

Theorem 2.5. *Let $\mathcal{A}$ be an abelian category with enough injectives and $F\colon \mathcal{A} \to \mathcal{B}$ an additive left exact functor. Then:*

1. *For each $n \geq 0$, R^nF is an additive functor $\mathcal{A} \to \mathcal{B}$, independent of the choice of injective resolutions up to unique natural isomorphism.*
2. *There is a natural isomorphism $F \cong R^0F$.*
3. *For each short exact sequence $0 \to M' \to M \to M'' \to 0$ and each $n \geq 0$ there is a connecting homomorphism $\delta^n\colon R^nF(M'') \to R^{n+1}F(M')$ giving a long exact sequence*

$$\cdots \to R^nF(M') \to R^nF(M) \to R^nF(M'') \xrightarrow{\delta^n} R^{n+1}F(M') \to R^{n+1}F(M) \to \cdots.$$

(4) *The connecting homomorphisms are natural: a morphism of short exact sequences from* $0 \to M' \to M \to M'' \to 0$ *to* $0 \to N' \to N \to N'' \to 0$*, with components* f, g, h*, yields commutative squares*

$$\begin{array}{ccc} R^nF(M'') & \xrightarrow{\delta^n} & R^{n+1}F(M') \\ {\scriptstyle R^nF(h)}\downarrow & & \downarrow{\scriptstyle R^{n+1}F(f)} \\ R^nF(N'') & \xrightarrow{\delta^n} & R^{n+1}F(N'). \end{array}$$

(5) *If* I *is injective, then* $R^nF(I) = 0$ *for all* $n > 0$.

Property (5) says that injective objects are invisible to the higher derived functors. Objects with this property play a central role in what follows, so we give them a name.

Definition 2.6. An object B of $\mathcal{B}$ is *G-acyclic* (for a left exact functor G on $\mathcal{B}$) if $R^qG(B) = 0$ for all $q > 0$.

The hypothesis of the Grothendieck spectral sequence will be that F sends injectives to G-acyclic objects. The most common way to verify this is the following lemma, which we will use repeatedly in Section 5.

Lemma 2.7. *Let* $F\colon \mathcal{A} \to \mathcal{B}$ *be an additive functor with an exact left adjoint* $L\colon \mathcal{B} \to \mathcal{A}$*. Then* F *sends injective objects to injective objects; in particular, it sends injectives to* G*-acyclic objects for every left exact* G *on* $\mathcal{B}$*.*

Proof. For I injective in $\mathcal{A}$, adjunction gives a natural isomorphism

$$\mathrm{Hom}_{\mathcal{B}}(-, F(I)) \cong \mathrm{Hom}_{\mathcal{A}}(L(-), I).$$

The right-hand side is the composite of the exact functor L with the exact functor $\mathrm{Hom}_{\mathcal{A}}(-, I)$, hence exact. The final statement follows from Theorem 2.5(5). □

3. Spectral Sequences

We now review the spectral sequence of a filtered complex, which is the only machinery the construction requires. We include a full proof, since the bookkeeping is where most of the confusion around spectral sequences lives. Standard references are [Wei94, Ch. 5] and [Lan02, §XX.9].

3.1. Definitions.

Definition 3.1. A (cohomological) *spectral sequence* in an abelian category consists of

- objects $E_r^{p,q}$ for $r \geq r_0$ and $p, q \in \mathbb{Z}$ (the r-th *page*),
- morphisms $d_r\colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$ with $d_r \circ d_r = 0$,
- isomorphisms $E_{r+1}^{p,q} \cong \ker\big(d_r\colon E_r^{p,q} \to E_r^{p+r,q-r+1}\big) / \operatorname{im}\big(d_r\colon E_r^{p-r,q+r-1} \to E_r^{p,q}\big)$.

In words: each page is a complex, and the next page is its cohomology.

The spectral sequence is *first quadrant* if $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$. In that case, for fixed (p, q) and $r > \max(p,\, q+1)$, the differential leaving $E_r^{p,q}$ lands in the region $q < 0$ and the differential arriving comes from the region $p < 0$, so both vanish and $E_r^{p,q} = E_{r+1}^{p,q} = \cdots$. This stable value is denoted $E_\infty^{p,q}$.

Definition 3.2. A first quadrant spectral sequence *converges* to a sequence of objects H^n, written

$$E_2^{p,q} \Longrightarrow H^{p+q},$$

if each H^n has a finite decreasing filtration $H^n = F^0H^n \supseteq F^1H^n \supseteq \cdots \supseteq F^{n+1}H^n = 0$ with $F^pH^n/F^{p+1}H^n \cong E_\infty^{p,n-p}$.

So the E_∞ page records the successive quotients of a filtration of H^n: the objects $E_\infty^{p,q}$ along the antidiagonal $p+q=n$ are the composition factors of H^n. Two consequences will be used often. First, the bottom row is a quotient of E_2 and a subobject of H^n, and the left column is a quotient of H^n and a subobject of E_2:

$$E_2^{n,0} \twoheadrightarrow E_\infty^{n,0} = F^nH^n \hookrightarrow H^n, \qquad H^n \twoheadrightarrow E_\infty^{0,n} \hookrightarrow E_2^{0,n}. \tag{3.1}$$

These are the *edge maps*. Second, over a field,

$$\dim H^n = \sum_{p+q=n} \dim E_\infty^{p,q} \leq \sum_{p+q=n} \dim E_2^{p,q};$$

counting dimensions this way is a surprisingly effective way of forcing differentials, as we will see in Section 6.

3.2. **The spectral sequence of a filtered complex.**

Definition 3.3. A *filtered complex* is a complex $(K^\bullet, d)$ together with a decreasing sequence of subcomplexes $K = F^0K \supseteq F^1K \supseteq F^2K \supseteq \cdots$. The filtration is *bounded* if $F^{n+1}K^n = 0$ for every n. The *associated graded* complexes are $\mathrm{Gr}^p K = F^pK/F^{p+1}K$, with differential induced by d.

The filtration on K induces one on cohomology: $F^pH^n(K)$ is the image of $H^n(F^pK) \to H^n(K)$, i.e., the classes that have a representative in F^pK. Writing Z^n and B^n for the cocycles and coboundaries of K,

$$F^pH^n(K) = \frac{Z^n \cap F^pK + B^n}{B^n}.$$

The guiding question is how the cohomology of the pieces $\mathrm{Gr}^p K$ (which is usually computable) relates to the cohomology of K (which is what we want). The spectral sequence answers it by successive approximation. To set it up, define for $r \geq 0$

$$Z_r^p = \{x \in F^pK : dx \in F^{p+r}K\},$$

the elements of filtration p that are cocycles "up to an error of filtration $p+r$". We also set $Z_{-1}^p = F^pK$. Each Z_r^p is graded by cohomological degree, and we write $Z_r^{p,q}$ for its part in degree $p+q$. Note the evident inclusions

$$Z_{r+1}^p \subseteq Z_r^p, \qquad Z_{r-1}^{p+1} \subseteq Z_r^p, \qquad Z_{r+1}^p \cap Z_{r-1}^{p+1} = Z_r^{p+1}, \qquad dZ_{r-1}^{p-r+1} \subseteq dZ_r^{p-r}. \tag{3.2}$$

(For the third: an element of $F^{p+1}K$ with $dx \in F^{p+r+1}K$ is exactly an element of Z_r^{p+1}.)

Theorem 3.4. *Let K be a filtered complex and, for $r \geq 0$, set*

$$E_r^{p,q} = \frac{Z_r^{p,q}}{Z_{r-1}^{p+1,q-1} + dZ_{r-1}^{p-r+1,\,q+r-2}}.$$

Then d induces maps $d_r \colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$ making $\{E_r, d_r\}$ a spectral sequence, with

$$E_0^{p,q} = \mathrm{Gr}^p K^{p+q}, \qquad E_1^{p,q} = H^{p+q}(\mathrm{Gr}^p K).$$

If the filtration is bounded and $F^0K = K$, the spectral sequence is first quadrant and $E_\infty^{p,q} \cong F^pH^{p+q}(K)/F^{p+1}H^{p+q}(K)$; that is, $E_1^{p,q} = H^{p+q}(\mathrm{Gr}^p K) \Longrightarrow H^{p+q}(K)$.

To lighten notation in the proof we suppress the second index q; every equality below is to be read degreewise. With this convention

$$E_r^p = Z_r^p / \big(Z_{r-1}^{p+1} + dZ_{r-1}^{p-r+1}\big).$$

Proof. The first two pages. For $r = 0$ we have $Z_0^p = F^pK$, and the denominator is $F^{p+1}K + dF^{p+1}K = F^{p+1}K$, so $E_0^p = \mathrm{Gr}^p K$. For $r = 1$, the numerator Z_1^p consists of elements of F^pK whose differential lies in $F^{p+1}K$; these are precisely lifts of cocycles of $\mathrm{Gr}^p K$. The denominator $F^{p+1}K + dF^pK$ consists of lifts of coboundaries of $\mathrm{Gr}^p K$. Hence $E_1^p = H(\mathrm{Gr}^p K)$.

The differential. If $x \in Z_r^p$ then $dx \in F^{p+r}K$ and $d(dx) = 0$, so $dx \in Z_r^{p+r}$. Moreover d carries the denominator of E_r^p into the denominator of E_r^{p+r}: it sends Z_{r-1}^{p+1} to $dZ_{r-1}^{p+1} = dZ_{r-1}^{(p+r)-r+1}$, and it kills dZ_{r-1}^{p-r+1}. So d induces $d_r \colon E_r^p \to E_r^{p+r}$, and $d_r^2 = 0$ because $d^2 = 0$.

Cycles of d_r. We claim

$$\ker\big(d_r \colon E_r^p \to E_r^{p+r}\big) = \frac{Z_{r+1}^p + Z_{r-1}^{p+1}}{Z_{r-1}^{p+1} + dZ_{r-1}^{p-r+1}}.$$

Suppose $x \in Z_r^p$ and $d_r[x] = 0$, i.e., $dx = z + dy$ with $z \in Z_{r-1}^{p+r+1}$ and $y \in Z_{r-1}^{p+1}$. Then $x - y \in F^pK$ and $d(x-y) = z \in F^{p+r+1}K$, so $x - y \in Z_{r+1}^p$ and $x \in Z_{r+1}^p + Z_{r-1}^{p+1}$. Conversely, if $x = u + y$ with $u \in Z_{r+1}^p$ and $y \in Z_{r-1}^{p+1}$, then du lies in $F^{p+r+1}K$ and is a cocycle, so $du \in Z_{r-1}^{p+r+1}$, while $dy \in dZ_{r-1}^{p+1}$; thus dx lies in the denominator of E_r^{p+r} and $d_r[x] = 0$. (Both Z_{r+1}^p and Z_{r-1}^{p+1} lie in Z_r^p by (3.2), so the displayed quotient makes sense.)

Boundaries of d_r. The image of $d_r \colon E_r^{p-r} \to E_r^p$ is represented by dZ_r^{p-r}, so

$$\mathrm{im}\big(d_r \colon E_r^{p-r} \to E_r^p\big) = \frac{dZ_r^{p-r} + Z_{r-1}^{p+1}}{Z_{r-1}^{p+1} + dZ_{r-1}^{p-r+1}},$$

where we used $dZ_{r-1}^{p-r+1} \subseteq dZ_r^{p-r}$ from (3.2).

Cohomology of E_r. Dividing, and using the second isomorphism theorem,

$$H(E_r)^p \cong \frac{Z_{r+1}^p + Z_{r-1}^{p+1}}{dZ_r^{p-r} + Z_{r-1}^{p+1}} \cong \frac{Z_{r+1}^p}{Z_{r+1}^p \cap \big(dZ_r^{p-r} + Z_{r-1}^{p+1}\big)}.$$

Since $dZ_r^{p-r} \subseteq F^pK$ consists of cocycles, it is contained in Z_{r+1}^p, so by the modular law and (3.2)

$$Z_{r+1}^p \cap \big(dZ_r^{p-r} + Z_{r-1}^{p+1}\big) = dZ_r^{p-r} + \big(Z_{r+1}^p \cap Z_{r-1}^{p+1}\big) = dZ_r^{p-r} + Z_r^{p+1}.$$

Therefore $H(E_r)^p \cong Z_{r+1}^p/(Z_r^{p+1} + dZ_r^{p-r}) = E_{r+1}^p$, which is exactly the defining formula for the next page.

Convergence. Assume $F^0K = K$ and $F^{n+1}K^n = 0$. Then $E_0^{p,q} = 0$ unless $p \geq 0$ and $q \geq 0$, hence the same holds on every page. Fix p and $n = p + q$, and take $r \geq \max(p,\, n + 2 - p)$. Then $F^{p+r}K^{n+1} = 0$ and $F^{p-r}K^{n-1} = K^{n-1}$, so

$$Z_r^p = Z^n \cap F^pK, \qquad Z_{r-1}^{p+1} = Z^n \cap F^{p+1}K, \qquad dZ_{r-1}^{p-r+1} = B^n \cap F^pK,$$

(the last because every $x \in K^{n-1}$ with $dx \in F^pK$ lies in Z_{r-1}^{p-r+1} once $p - r + 1 \leq 0$). Hence

$$E_\infty^{p,q} = \frac{Z^n \cap F^pK}{Z^n \cap F^{p+1}K + B^n \cap F^pK}.$$

On the other hand, by the second isomorphism theorem and the modular law,

$$\frac{F^pH^n}{F^{p+1}H^n} = \frac{Z^n \cap F^pK + B^n}{Z^n \cap F^{p+1}K + B^n} \cong \frac{Z^n \cap F^pK}{Z^n \cap F^{p+1}K + B^n \cap F^pK},$$

which is the same object. Finally the filtration of H^n satisfies $F^0H^n = H^n$ and $F^{n+1}H^n = 0$, as required. □

Remark 3.5. The first differential has a familiar description: the map $d_1 \colon H^n(\mathrm{Gr}^p K) \to H^{n+1}(\mathrm{Gr}^{p+1} K)$ is the connecting homomorphism of the short exact sequence of complexes

$$0 \to \mathrm{Gr}^{p+1} K \to F^pK/F^{p+2}K \to \mathrm{Gr}^p K \to 0.$$

Indeed, the connecting map is computed by lifting a cocycle of $\mathrm{Gr}^p K$ to $x \in F^pK$ and taking the class of dx, which is exactly how d_1 is defined. In the same spirit, d_r "lifts a class $r-1$ more steps and then differentiates".

3.3. **Double complexes.** The filtered complexes we need come from double complexes.

Definition 3.6. A *first quadrant double complex* K is a collection of objects $K^{p,q}$ $(p, q \geq 0)$ with a horizontal differential $d_h \colon K^{p,q} \to K^{p+1,q}$ and a vertical differential $d_v \colon K^{p,q} \to K^{p,q+1}$ satisfying $d_h^2 = 0$, $d_v^2 = 0$, and $d_h d_v = d_v d_h$. Its *total complex* is

$$\mathrm{Tot}(K)^n = \bigoplus_{p+q=n} K^{p,q}, \qquad d = d_h + (-1)^p d_v \ \text{ on } K^{p,q}.$$

The sign ensures $d^2 = 0$.

We write $H_v^q(K^{p,\bullet})$ for the cohomology of the p-th column with respect to d_v, and $H_h^p(K^{\bullet,q})$ for the cohomology of the q-th row with respect to d_h. Because the differentials commute, d_h induces a differential on the objects $H_v^q(K^{p,\bullet})$ for varying p, and d_v induces one on $H_h^p(K^{\bullet,q})$ for varying q. We denote the resulting iterated cohomologies by $H_h^pH_v^q(K)$ and $H_v^qH_h^p(K)$.

The total complex carries two natural bounded filtrations, by columns and by rows:

$${}^{I}F^p \mathrm{Tot}(K)^n = \bigoplus_{i \geq p} K^{i,n-i}, \qquad {}^{II}F^p \mathrm{Tot}(K)^n = \bigoplus_{j \geq p} K^{n-j,j}.$$

Proposition 3.7. *Let K be a first quadrant double complex. There are two first quadrant spectral sequences converging to $H^{p+q}(\mathrm{Tot}\, K)$:*

$${}^{I}E_2^{p,q} = H_h^pH_v^q(K) \Longrightarrow H^{p+q}(\mathrm{Tot}\, K), \qquad {}^{II}E_2^{p,q} = H_v^pH_h^q(K) \Longrightarrow H^{p+q}(\mathrm{Tot}\, K).$$

Proof. Apply Theorem 3.4 to the column filtration. Its associated graded in degree $p+q$ is $\mathrm{Gr}^p \mathrm{Tot}(K)^{p+q} = K^{p,q}$, with differential $\pm d_v$, so ${}^{I}E_1^{p,q} = H_v^q(K^{p,\bullet})$ (the sign does not affect cohomology).

To compute d_1, note that every class in ${}^{I}E_1^{p,q}$ is represented by some $x \in K^{p,q}$ with $d_v x = 0$: a general representative lies in $F^p \mathrm{Tot}$, and its components in columns $> p$ lie in Z_0^{p+1}, which is part of the denominator. For such x we have $dx = d_h x \in K^{p+1,q}$, so $d_1[x] = [d_h x]$. Thus d_1 is the map induced by d_h on vertical cohomology, and ${}^{I}E_2^{p,q} = H_h^pH_v^q(K)$.

The row filtration is the column filtration of the transposed double complex K^{T}, with $(K^{\mathrm{T}})^{p,q} = K^{q,p}$ and the roles of d_h, d_v exchanged. The total complexes of K and K^{T} are isomorphic (via the sign $(-1)^{pq}$ on $K^{p,q}$), and this isomorphism respects the filtrations, so the first argument gives ${}^{II}E_2^{p,q} = H_v^pH_h^q(K)$. Both filtrations are bounded, so both spectral sequences converge to $H^\bullet(\mathrm{Tot}\, K)$. □

The strategy behind the Grothendieck spectral sequence, and many other spectral sequence arguments, is now visible: build a double complex for which one of the two spectral sequences is trivial to evaluate. That spectral sequence then *computes* the abutment $H^\bullet(\operatorname{Tot} K)$, and the other spectral sequence becomes a nontrivial statement about it.

3.4. **Low-degree consequences.** Even when a spectral sequence cannot be computed completely, its low-degree terms often can.

Proposition 3.8 (Five-term exact sequence)**.** *For any first quadrant spectral sequence $E_2^{p,q} \Rightarrow H^{p+q}$ there is an exact sequence*

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2.$$

Proof. All differentials into and out of $E_r^{1,0}$ for $r \geq 2$ start or end outside the first quadrant, so $E_\infty^{1,0} = E_2^{1,0}$. For $E^{0,1}$, the only possibly nonzero differential is $d_2 \colon E_2^{0,1} \to E_2^{2,0}$ (later ones land in negative q), so $E_\infty^{0,1} = \ker d_2$. For $E^{2,0}$, the only possibly nonzero incoming differential is this same d_2, so $E_\infty^{2,0} = \operatorname{coker} d_2$. Now the filtration of H^1 has two steps, giving the short exact sequence $0 \to E_\infty^{1,0} \to H^1 \to E_\infty^{0,1} \to 0$, and $E_\infty^{2,0} = F^2H^2 \subseteq H^2$. Splicing these together with the exact sequence $0 \to \ker d_2 \to E_2^{0,1} \to E_2^{2,0} \to \operatorname{coker} d_2 \to 0$ gives the result. □

Proposition 3.9. *If a first quadrant spectral sequence $E_2^{p,q} \Rightarrow H^{p+q}$ has $E_2^{p,q} = 0$ for all $q \neq 0$, then $H^n \cong E_2^{n,0}$. Likewise, if $E_2^{p,q} = 0$ for all $p \neq 0$, then $H^n \cong E_2^{0,n}$.*

Proof. In either case every d_r with $r \geq 2$ has source or target zero, so $E_2 = E_\infty$. Each antidiagonal then contains a single nonzero term, so the filtration of H^n has only one nonzero graded piece. □

When the hypothesis of Proposition 3.9 holds we say the spectral sequence *collapses*.

4. The Grothendieck Spectral Sequence

To apply the double complex strategy to a composite $G \circ F$, we need a resolution of a *complex* $F(C^\bullet)$ by injectives that is fine enough to resolve its cocycles, coboundaries, and cohomology simultaneously. This is a Cartan–Eilenberg resolution.

4.1. **Cartan–Eilenberg resolutions.** For a complex $A^\bullet$ we write $Z^p(A)$, $B^p(A)$, $H^p(A)$ for its cocycles, coboundaries, and cohomology in degree p. For a double complex I we use the same notation for the horizontal differential applied to a fixed row: $Z_h^p(I^{\bullet,q}) = \ker(d_h \colon I^{p,q} \to I^{p+1,q})$, and similarly B_h^p and H_h^p. Since d_v commutes with d_h, it maps each of these into itself as q varies.

Definition 4.1. Let $A^\bullet$ be a complex in $\mathcal{B}$ with $A^p = 0$ for $p < 0$. A *Cartan–Eilenberg resolution* of $A^\bullet$ is a first quadrant double complex $I^{\bullet,\bullet}$ of injective objects together with an augmentation $\varepsilon \colon A^p \to I^{p,0}$, compatible with the horizontal differentials, such that for every p:

(1) $0 \to A^p \to I^{p,0} \to I^{p,1} \to \cdots$ is an injective resolution of A^p;
(2) the induced sequences

$$0 \to Z^p(A) \to Z_h^p(I^{\bullet,0}) \to Z_h^p(I^{\bullet,1}) \to \cdots,$$

and the analogous ones for B^p and H^p, are injective resolutions of $Z^p(A)$, $B^p(A)$, and $H^p(A)$.

Pictorially, a Cartan–Eilenberg resolution of $A^\bullet$ is a double complex sitting above it,

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & I^{0,1} & \longrightarrow & I^{1,1} & \longrightarrow & I^{2,1} & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & I^{0,0} & \longrightarrow & I^{1,0} & \longrightarrow & I^{2,0} & \longrightarrow & \cdots \\
 & & \varepsilon\uparrow & & \varepsilon\uparrow & & \varepsilon\uparrow & & \\
0 & \longrightarrow & A^0 & \longrightarrow & A^1 & \longrightarrow & A^2 & \longrightarrow & \cdots \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

in which every column is an injective resolution, and moreover taking horizontal cocycles, coboundaries, or cohomology row by row produces injective resolutions of the corresponding pieces of $A^\bullet$.

Proposition 4.2 ([Wei94, Lemma 5.7.2])**.** *If $\mathcal{B}$ has enough injectives, every complex $A^\bullet$ in $\mathcal{B}$ with $A^p = 0$ for $p < 0$ has a Cartan–Eilenberg resolution.*

Sketch. For each p, choose injective resolutions $J_B^{p,\bullet}$ of $B^p(A)$ and $J_H^{p,\bullet}$ of $H^p(A)$. The horseshoe lemma applied to $0 \to B^p(A) \to Z^p(A) \to H^p(A) \to 0$ produces an injective resolution $J_Z^{p,\bullet}$ of $Z^p(A)$ fitting into a degreewise split exact sequence $0 \to J_B^{p,\bullet} \to J_Z^{p,\bullet} \to J_H^{p,\bullet} \to 0$. Applying the horseshoe lemma again to $0 \to Z^p(A) \to A^p \to B^{p+1}(A) \to 0$ produces an injective resolution $I^{p,\bullet}$ of A^p with $0 \to J_Z^{p,\bullet} \to I^{p,\bullet} \to J_B^{p+1,\bullet} \to 0$. Finally define d_h as the composite $I^{p,\bullet} \to J_B^{p+1,\bullet} \hookrightarrow J_Z^{p+1,\bullet} \hookrightarrow I^{p+1,\bullet}$. One checks directly that $Z_h^p = J_Z^{p,\bullet}$, $B_h^p = J_B^{p,\bullet}$, and $H_h^p = J_H^{p,\bullet}$. □

The key structural feature of a Cartan–Eilenberg resolution is that its rows are split, so that *any* additive functor commutes with taking their cohomology.

Lemma 4.3. *Let $J^\bullet$ be a complex such that $Z^p(J)$ and $B^p(J)$ are injective for every p, and let G be any additive functor. Then there are natural isomorphisms*

$$Z^p(GJ) \cong G\big(Z^p(J)\big), \qquad B^p(GJ) \cong G\big(B^p(J)\big), \qquad H^p(GJ) \cong G\big(H^p(J)\big).$$

Proof. Consider the two short exact sequences

$$0 \to Z^p(J) \to J^p \to B^{p+1}(J) \to 0, \qquad 0 \to B^p(J) \to Z^p(J) \to H^p(J) \to 0.$$

Both split, because their left-hand terms are injective. Since G is additive, it carries split exact sequences to split exact sequences. The differential $J^p \to J^{p+1}$ factors as

$$J^p \twoheadrightarrow B^{p+1}(J) \hookrightarrow Z^{p+1}(J) \hookrightarrow J^{p+1},$$

a split epimorphism followed by two split monomorphisms. Applying G, the composite $G(J^p) \to G(J^{p+1})$ is again a split epimorphism followed by split monomorphisms. Its kernel is therefore $\ker\big(G(J^p) \to G(B^{p+1}(J))\big) = G(Z^p(J))$, and its image is $G(B^{p+1}(J))$. This gives the first two isomorphisms, and the third follows by applying G to the split sequence $0 \to B^p \to Z^p \to H^p \to 0$. □

Remark 4.4. In a Cartan–Eilenberg resolution, every row $I^{\bullet,q}$ satisfies the hypothesis of Lemma 4.3: its horizontal cocycles and coboundaries are terms of injective resolutions, hence injective.

4.2. **The main theorem.**

Theorem 4.5 (Grothendieck spectral sequence)**.** *Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be abelian categories, and assume $\mathcal{A}$ and $\mathcal{B}$ have enough injectives. Let $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ be additive left exact functors, and suppose F sends injective objects of $\mathcal{A}$ to G-acyclic objects of $\mathcal{B}$. Then for every object A of $\mathcal{A}$ there is a first quadrant spectral sequence, natural in A from the E_2 page on,*

$$E_2^{p,q} = R^pG\big(R^qF(A)\big) \Longrightarrow R^{p+q}(G \circ F)(A).$$

Note that $G \circ F$ is left exact, so its right derived functors are defined.

Proof. Step 1: the double complex. Choose an injective resolution $A \to C^\bullet$ in $\mathcal{A}$. Applying F gives a complex $F(C^\bullet)$ in $\mathcal{B}$, concentrated in nonnegative degrees, whose cohomology is by definition

$$H^q\big(F(C^\bullet)\big) = R^qF(A). \tag{4.1}$$

By Proposition 4.2, choose a Cartan–Eilenberg resolution $F(C^\bullet) \to I^{\bullet,\bullet}$, and apply G termwise to obtain a first quadrant double complex $G(I)$ in $\mathcal{C}$. We will compute $H^n\big(\operatorname{Tot} G(I)\big)$ in two ways using Proposition 3.7.

Step 2: the column filtration collapses. The p-th column of I is an injective resolution of $F(C^p)$. Hence the p-th column of $G(I)$ computes the derived functors of G on $F(C^p)$:

$${}^{I}E_1^{p,q} = H_v^q\big(G(I^{p,\bullet})\big) = R^qG\big(F(C^p)\big).$$

Since C^p is injective, $F(C^p)$ is G-acyclic by hypothesis, so this vanishes for $q > 0$. For $q = 0$ we get $R^0G(F(C^p)) \cong G(F(C^p))$, and by naturality of this isomorphism d_1 is identified with $GF(d_C)$. Therefore

$${}^{I}E_2^{p,q} = \begin{cases} H^p\big(GF(C^\bullet)\big) = R^p(G \circ F)(A), & q = 0, \\ 0, & q > 0, \end{cases}$$

where we used that $C^\bullet$ is an injective resolution of A. This spectral sequence collapses (Proposition 3.9), so

$$H^n\big(\operatorname{Tot} G(I)\big) \cong R^n(G \circ F)(A). \tag{4.2}$$

Step 3: the row filtration is the Grothendieck spectral sequence. The second spectral sequence of $G(I)$ has

$${}^{II}E_1^{p,q} = H_h^q\big(G(I^{\bullet,p})\big) \cong G\big(H_h^q(I^{\bullet,p})\big),$$

where the isomorphism is Lemma 4.3 applied to the row $I^{\bullet,p}$. These isomorphisms are natural, so they are compatible with the maps induced by d_v, and the d_1 differential becomes G applied to the complex

$$H_h^q(I^{\bullet,0}) \to H_h^q(I^{\bullet,1}) \to H_h^q(I^{\bullet,2}) \to \cdots. \tag{4.3}$$

By the defining property of a Cartan–Eilenberg resolution, (4.3) is an injective resolution of $H^q(F(C^\bullet)) = R^qF(A)$, using (4.1). Taking cohomology of G applied to it computes the derived functors of G:

$${}^{II}E_2^{p,q} = H^p\Big(G\big(H_h^q(I^{\bullet,\bullet})\big)\Big) = R^pG\big(R^qF(A)\big).$$

By Proposition 3.7 this spectral sequence converges to $H^{p+q}(\operatorname{Tot} G(I))$, which is $R^{p+q}(G \circ F)(A)$ by (4.2).

Step 4: naturality. Any two injective resolutions of A are homotopy equivalent, and Cartan–Eilenberg resolutions are unique up to a homotopy that respects the column and row

filtrations [Wei94, §5.7]. A filtration-preserving homotopy induces the identity from the E_1 page of the row spectral sequence onward, so from E_2 on the spectral sequence is independent of all choices and functorial in A. □

Remark 4.6. It is worth isolating where each hypothesis was used. The G-acyclicity of F(injectives) made the first spectral sequence collapse, which identified the abutment. The Cartan–Eilenberg property, through Lemma 4.3, let G pass through horizontal cohomology, which identified the E_2 page. Left exactness of G gave $R^0G \cong G$ in Step 2.

Combining the theorem with Propositions 3.8 and 3.9 gives the following consequences, which are often all one needs.

Corollary 4.7. *In the situation of Theorem 4.5:*

(1) *There is an exact sequence*

$$0 \to R^1G(FA) \to R^1(GF)(A) \to G(R^1F(A)) \to R^2G(FA) \to R^2(GF)(A).$$

(2) *If $R^qF(A) = 0$ for all $q > 0$, then $R^n(GF)(A) \cong R^nG(F(A))$ for all n.*

(3) *If G is exact, then $R^n(GF)(A) \cong G(R^nF(A))$ for all n.*

Proof. Part (1) is Proposition 3.8, using $R^0F \cong F$ and $R^0G \cong G$. In part (2) the E_2 page is concentrated in the row $q = 0$; in part (3) it is concentrated in the column $p = 0$. Both then follow from Proposition 3.9. □

5. Three Classical Special Cases

In each case below, the hypothesis of Theorem 4.5 is verified by Lemma 2.7: the first functor is a right adjoint of an exact functor.

5.1. **The Leray spectral sequence.** Let $f\colon X \to Y$ be a continuous map of topological spaces. For a sheaf $\mathscr{F} \in \mathrm{Sh}(X)$, the *direct image* $f_*\mathscr{F} \in \mathrm{Sh}(Y)$ is defined by $(f_*\mathscr{F})(V) = \mathscr{F}(f^{-1}V)$. The functor $f_*\colon \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ is left exact, and its right derived functors R^qf_* are the *higher direct images*. Concretely, $R^qf_*\mathscr{F}$ is the sheaf associated to the presheaf $V \mapsto H^q(f^{-1}V, \mathscr{F})$ [Har77, Prop. III.8.1], so it measures the cohomology of $\mathscr{F}$ along the fibres of f.

Theorem 5.1 (Leray spectral sequence)**.** *For every sheaf $\mathscr{F}$ of abelian groups on X there is a spectral sequence*

$$E_2^{p,q} = H^p\big(Y, R^qf_*\mathscr{F}\big) \Longrightarrow H^{p+q}(X, \mathscr{F}).$$

Proof. Take $F = f_*$ and $G = \Gamma(Y, -)$, both left exact, between categories with enough injectives. The inverse image functor $f^{-1}\colon \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ is exact and left adjoint to f_*, so f_* preserves injectives by Lemma 2.7. For every sheaf $\mathscr{F}$ on X we have $\Gamma(Y, f_*\mathscr{F}) = \mathscr{F}(f^{-1}Y) = \Gamma(X, \mathscr{F})$, so $G \circ F = \Gamma(X, -)$ and $R^n(G \circ F) = H^n(X, -)$. Theorem 4.5 gives the result. □

The Leray spectral sequence approximates the cohomology of X by the cohomology of Y with coefficients in the higher direct images. The case used most often is the collapse of Corollary 4.7(2):

Corollary 5.2. *If $R^qf_*\mathscr{F} = 0$ for all $q > 0$, then $H^n(X, \mathscr{F}) \cong H^n(Y, f_*\mathscr{F})$ for all n.*

This applies, for example, when f is a closed embedding (then f_* is exact) or, in algebraic geometry, when f is an affine morphism of noetherian schemes and $\mathscr{F}$ is quasi-coherent. In the latter case $f^{-1}V$ is affine for every affine open $V \subseteq Y$, so $H^q(f^{-1}V, \mathscr{F}) = 0$ for $q > 0$ by Serre's vanishing theorem [Har77, Thm. III.3.5], and therefore $R^qf_*\mathscr{F} = 0$.

5.2. **The Lyndon–Hochschild–Serre spectral sequence.** Let Π be a group, $N \trianglelefteq \Pi$ a normal subgroup, and $Q = \Pi/N$. Group cohomology is $H^n(\Pi, M) = R^n(-)^{\Pi}(M)$, the derived functors of invariants on the category Mod_Π of Π-modules [Bro82, Ch. III].

Theorem 5.3 (Lyndon–Hochschild–Serre)**.** *For every Π-module M there is a spectral sequence*

$$E_2^{p,q} = H^p\big(Q,\, H^q(N, M)\big) \Longrightarrow H^{p+q}(\Pi, M).$$

Proof. If M is a Π-module, the invariants M^N form a Q-module, giving a left exact functor $F = (-)^N\colon \mathrm{Mod}_\Pi \to \mathrm{Mod}_Q$. Let $G = (-)^Q\colon \mathrm{Mod}_Q \to \mathbf{Ab}$. Then $G \circ F = (-)^\Pi$. The functor $(-)^N$ is right adjoint to the inflation functor $\mathrm{Mod}_Q \to \mathrm{Mod}_\Pi$ (regard a Q-module as a Π-module via $\Pi \to Q$), which is exact, so F preserves injectives by Lemma 2.7.

It remains to identify $R^qF(M)$. The underlying abelian group of $R^qF(M)$ is computed by applying $(-)^N$ to an injective resolution of M in Mod_Π. The restriction functor $\mathrm{Mod}_\Pi \to \mathrm{Mod}_N$ is exact and preserves injectives (it has the exact left adjoint $\mathbb{Z}[\Pi] \otimes_{\mathbb{Z}[N]} -$, exact because $\mathbb{Z}[\Pi]$ is free over $\mathbb{Z}[N]$), so that resolution is also an injective resolution in Mod_N. Hence $R^qF(M) = H^q(N, M)$, with Q acting through conjugation [Bro82, §III.8]. □

The five-term exact sequence of Corollary 4.7(1) becomes the classical *inflation–restriction* sequence

$$0 \to H^1(Q, M^N) \to H^1(\Pi, M) \to H^1(N, M)^Q \to H^2(Q, M^N) \to H^2(\Pi, M). \tag{5.1}$$

5.3. **A change-of-rings spectral sequence.** Let $\varphi\colon R \to S$ be a homomorphism of commutative rings. Restriction of scalars $\mathrm{Mod}_S \to \mathrm{Mod}_R$ is exact, and its right adjoint is $\mathrm{Hom}_R(S, -)\colon \mathrm{Mod}_R \to \mathrm{Mod}_S$, where S acts through the first variable.

Theorem 5.4. *For an S-module N and an R-module M there is a spectral sequence*

$$E_2^{p,q} = \mathrm{Ext}_S^p\big(N, \mathrm{Ext}_R^q(S, M)\big) \Longrightarrow \mathrm{Ext}_R^{p+q}(N, M).$$

Proof. Let $F = \mathrm{Hom}_R(S, -)$ and $G = \mathrm{Hom}_S(N, -)$. By Lemma 2.7, F preserves injectives. By the tensor–hom adjunction, $G \circ F = \mathrm{Hom}_S(N, \mathrm{Hom}_R(S, -)) \cong \mathrm{Hom}_R(N \otimes_S S, -) = \mathrm{Hom}_R(N, -)$. Finally $R^qF(M)$ is computed by applying $\mathrm{Hom}_R(S, -)$ to an injective resolution of M, so $R^qF(M) = \mathrm{Ext}_R^q(S, M)$ with its natural S-module structure. □

6. Example Computations

We now put the spectral sequences of Section 5 to work. The first example uses a collapsing Leray spectral sequence as a black box to prove something concrete. The other two have nonzero differentials, and illustrate how differentials can be forced by knowing the abutment.

6.1. Pushing forward the structure sheaf of $\mathbb{P}^1$.

Example 6.1. Let k be a field, $d \geq 1$, and let $f\colon \mathbb{P}^1_k \to \mathbb{P}^1_k$ be the morphism $[x : y] \mapsto [x^d : y^d]$. We determine the vector bundle $f_*\mathscr{O}_{\mathbb{P}^1}$.

The map f is finite, in particular affine. Over the open set $U = \{x \neq 0\}$ with coordinate $t = y/x$ on the target and $s = y/x$ on the source, f corresponds to $k[t] \to k[s]$, $t \mapsto s^d$, and $k[s]$ is free over $k[t]$ with basis $1, s, \ldots, s^{d-1}$; the same holds over $\{y \neq 0\}$. So $f_*\mathscr{O}_{\mathbb{P}^1}$ is locally free of rank d. By Grothendieck's splitting theorem it decomposes as

$$f_*\mathscr{O}_{\mathbb{P}^1} \cong \bigoplus_{i=1}^{d} \mathscr{O}(a_i)$$

for some integers a_i.

Since f is affine and $\mathscr{O}_{\mathbb{P}^1}$ is quasi-coherent, $R^q f_* \mathscr{O}_{\mathbb{P}^1} = 0$ for $q > 0$, and the Leray spectral sequence collapses (Corollary 5.2):

$$H^n\big(\mathbb{P}^1, f_*\mathscr{O}_{\mathbb{P}^1}\big) \cong H^n\big(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}\big) = \begin{cases} k, & n = 0, \\ 0, & n \geq 1. \end{cases}$$

Recall that $h^0(\mathscr{O}(a)) = \max(a+1, 0)$ and $h^1(\mathscr{O}(a)) = \max(-a-1, 0)$ [Har77, Thm. III.5.1]. The vanishing of H^1 forces $a_i \geq -1$ for every i. Then $h^0 = \sum_i (a_i + 1) = 1$ is a sum of nonnegative integers, so exactly one a_i equals 0 and the rest equal -1. Hence

$$f_*\mathscr{O}_{\mathbb{P}^1} \cong \mathscr{O} \oplus \mathscr{O}(-1)^{\oplus(d-1)}.$$

One can confirm this by hand: the summand $\mathscr{O}$ is spanned by 1, and the summands $\mathscr{O}(-1)$ by $s, \dots, s^{d-1}$, which acquire poles at infinity. The point is that the spectral sequence argument required no such bookkeeping.

6.2. **A nonzero d_2 in group cohomology.** We use the Lyndon–Hochschild–Serre spectral sequence to compare two groups that it cannot distinguish on the E_2 page. Let C_m denote the cyclic group of order m with generator t, and recall the periodic free resolution of the trivial module $\mathbb{Z}$ over $\mathbb{Z}[C_m]$ [Bro82, §I.6]:

$$\cdots \xrightarrow{N} \mathbb{Z}[C_m] \xrightarrow{t-1} \mathbb{Z}[C_m] \xrightarrow{N} \mathbb{Z}[C_m] \xrightarrow{t-1} \mathbb{Z}[C_m] \to \mathbb{Z} \to 0, \qquad N = 1 + t + \cdots + t^{m-1}.$$

Applying $\mathrm{Hom}_{C_m}(-, \mathbb{F}_2)$ with trivial action, $t - 1$ acts as 0 and N acts as multiplication by m. When m is even all differentials vanish, so

$$H^n(C_m, \mathbb{F}_2) \cong \mathbb{F}_2 \qquad \text{for all } n \geq 0 \quad (m \text{ even}). \tag{6.1}$$

Example 6.2. Let Π be either C_4 or $C_2 \times C_2$, let $N \cong C_2$ be the subgroup $\{0, 2\} \subset C_4$ or $C_2 \times 0 \subset C_2 \times C_2$ respectively, and let $M = \mathbb{F}_2$ with trivial action. In both cases $Q = \Pi/N \cong C_2$. Since Π is abelian and M is trivial, Q acts trivially on $H^q(N, \mathbb{F}_2)$, so by (6.1) both groups have the *same* E_2 page:

$$E_2^{p,q} = H^p\big(C_2, H^q(C_2, \mathbb{F}_2)\big) \cong \mathbb{F}_2 \qquad \text{for all } p, q \geq 0.$$

The antidiagonal $p + q = n$ has $n + 1$ copies of $\mathbb{F}_2$. The abutments, however, differ:

- $H^n(C_2 \times C_2, \mathbb{F}_2) \cong \mathbb{F}_2^{n+1}$ by the Künneth formula;
- $H^n(C_4, \mathbb{F}_2) \cong \mathbb{F}_2$ by (6.1).

The split case. For $\Pi = C_2 \times C_2$, we have $\dim H^n = n + 1 = \sum_{p+q=n} \dim E_2^{p,q}$. Since $E_\infty^{p,q}$ is a subquotient of $E_2^{p,q}$, the inequality $\dim H^n \leq \sum \dim E_2^{p,q}$ is an equality only if $E_\infty = E_2$. So every differential vanishes and the spectral sequence collapses at E_2.

The non-split case. For $\Pi = C_4$, the five-term sequence (5.1) reads

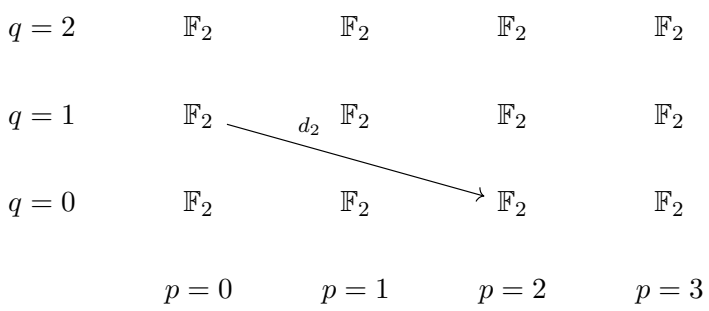
$$0 \to \underbrace{H^1(C_2, \mathbb{F}_2)}_{\mathbb{F}_2} \to \underbrace{H^1(C_4, \mathbb{F}_2)}_{\mathbb{F}_2} \to \underbrace{H^1(C_2, \mathbb{F}_2)}_{\mathbb{F}_2} \xrightarrow{d_2} \underbrace{H^2(C_2, \mathbb{F}_2)}_{\mathbb{F}_2} \to \underbrace{H^2(C_4, \mathbb{F}_2)}_{\mathbb{F}_2}.$$

The first map is an injection between one-dimensional spaces, hence an isomorphism. The next map is therefore zero, so d_2 is injective, and hence

$$d_2 \colon E_2^{0,1} \xrightarrow{\cong} E_2^{2,0}.$$

So the differential d_2 is nonzero for C_4 and zero for $C_2 \times C_2$. In other words, the spectral sequence sees through the identical E_2 pages and detects that $0 \to C_2 \to C_4 \to C_2 \to 0$ is a

non-split extension. In fact $d_2^{0,1}$ sends the generator of $H^1(N, \mathbb{F}_2)$ to the class in $H^2(Q, \mathbb{F}_2)$ classifying the extension, which is nonzero precisely when the extension does not split.



Remark 6.3. Dimension counting alone does not determine the rest of the spectral sequence for C_4. Using the multiplicative structure (the E_2 page is the polynomial ring $\mathbb{F}_2[x, y]$ with $x \in E_2^{1,0}$, $y \in E_2^{0,1}$, and d_2 is a derivation with $d_2 x = 0$, $d_2 y = x^2$), one finds $E_3 = E_\infty \cong \mathbb{F}_2[x]/(x^2) \otimes \mathbb{F}_2[y^2]$. This has exactly one class in each total degree, in agreement with (6.1). See [Bro82, §VII.6] or [Wei94, §6.8].

6.3. **Differentials forced by the abutment.** Our last example shows a spectral sequence in which infinitely many differentials are nonzero, all of them forced by knowledge of the abutment.

Example 6.4. Apply Theorem 5.4 with $R = \mathbb{Z}$, $S = \mathbb{Z}/4$, $N = \mathbb{Z}/2$ (as a $\mathbb{Z}/4$-module), and $M = \mathbb{Z}/2$:

$$E_2^{p,q} = \operatorname{Ext}^p_{\mathbb{Z}/4}\big(\mathbb{Z}/2,\, \operatorname{Ext}^q_{\mathbb{Z}}(\mathbb{Z}/4, \mathbb{Z}/2)\big) \Longrightarrow \operatorname{Ext}^{p+q}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2).$$

The coefficients. From the resolution $0 \to \mathbb{Z} \xrightarrow{4} \mathbb{Z} \to \mathbb{Z}/4 \to 0$, applying $\operatorname{Hom}_{\mathbb{Z}}(-, \mathbb{Z}/2)$ gives $\mathbb{Z}/2 \xrightarrow{4=0} \mathbb{Z}/2$, so $\operatorname{Ext}^q_{\mathbb{Z}}(\mathbb{Z}/4, \mathbb{Z}/2) \cong \mathbb{Z}/2$ for $q = 0, 1$ and vanishes for $q \geq 2$. (The only $\mathbb{Z}/4$-module structure on $\mathbb{Z}/2$ is the obvious one.)

The E_2 page. Over $\mathbb{Z}/4$, the module $\mathbb{Z}/2$ has the periodic free resolution

$$\cdots \xrightarrow{2} \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \xrightarrow{2} \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0.$$

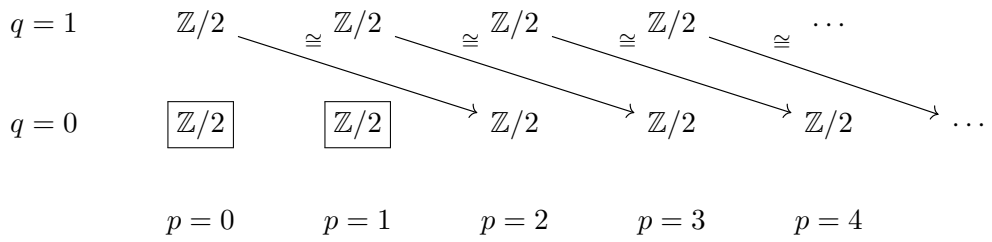
Applying $\operatorname{Hom}_{\mathbb{Z}/4}(-, \mathbb{Z}/2)$, every map becomes multiplication by 2 on $\mathbb{Z}/2$, which is zero. So $\operatorname{Ext}^p_{\mathbb{Z}/4}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$ for every $p \geq 0$, and the E_2 page has two nonzero rows:

$$E_2^{p,q} \cong \begin{cases} \mathbb{Z}/2, & p \geq 0,\ q \in \{0, 1\}, \\ 0, & \text{otherwise.} \end{cases}$$

The abutment. Since $\mathbb{Z}$ is a principal ideal domain, every $\mathbb{Z}$-module has an injective resolution of length one, so $\operatorname{Ext}^n_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = 0$ for $n \geq 2$. Directly, $\operatorname{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$.

The differentials. With only two rows, the only possibly nonzero differential is $d_2 \colon E_2^{p,1} \to E_2^{p+2,0}$, and $E_3 = E_\infty$. For $p \geq 0$, the group $E_\infty^{p+2,0} = \operatorname{coker}\big(d_2 \colon E_2^{p,1} \to E_2^{p+2,0}\big)$ is a subobject of $\operatorname{Ext}^{p+2}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = 0$. So every such d_2 is surjective, and being a map $\mathbb{Z}/2 \to \mathbb{Z}/2$, it is

an isomorphism:



Consequently $E_\infty^{p,1} = \ker d_2 = 0$ for all p, and the only survivors are the boxed terms $E_\infty^{0,0}$ and $E_\infty^{1,0}$. They recover $\mathrm{Ext}^0_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$ and $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$, consistent with the five-term sequence

$$0 \to \underbrace{E_2^{1,0}}_{\mathbb{Z}/2} \xrightarrow{\cong} \underbrace{\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2)}_{\mathbb{Z}/2} \xrightarrow{0} \underbrace{E_2^{0,1}}_{\mathbb{Z}/2} \xrightarrow{\cong} \underbrace{E_2^{2,0}}_{\mathbb{Z}/2} \to 0.$$

The infinite tail of the E_2 page reflects the fact that $\mathbb{Z}/2$ has infinite projective dimension over $\mathbb{Z}/4$. It must cancel against itself, because $\mathbb{Z}/2$ has projective dimension one over $\mathbb{Z}$, and the differentials d_2 are exactly how it does.

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